

Explain Like I'm 5

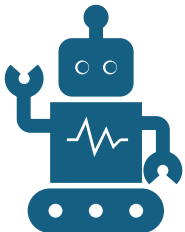
AI - powered multi-level training tool

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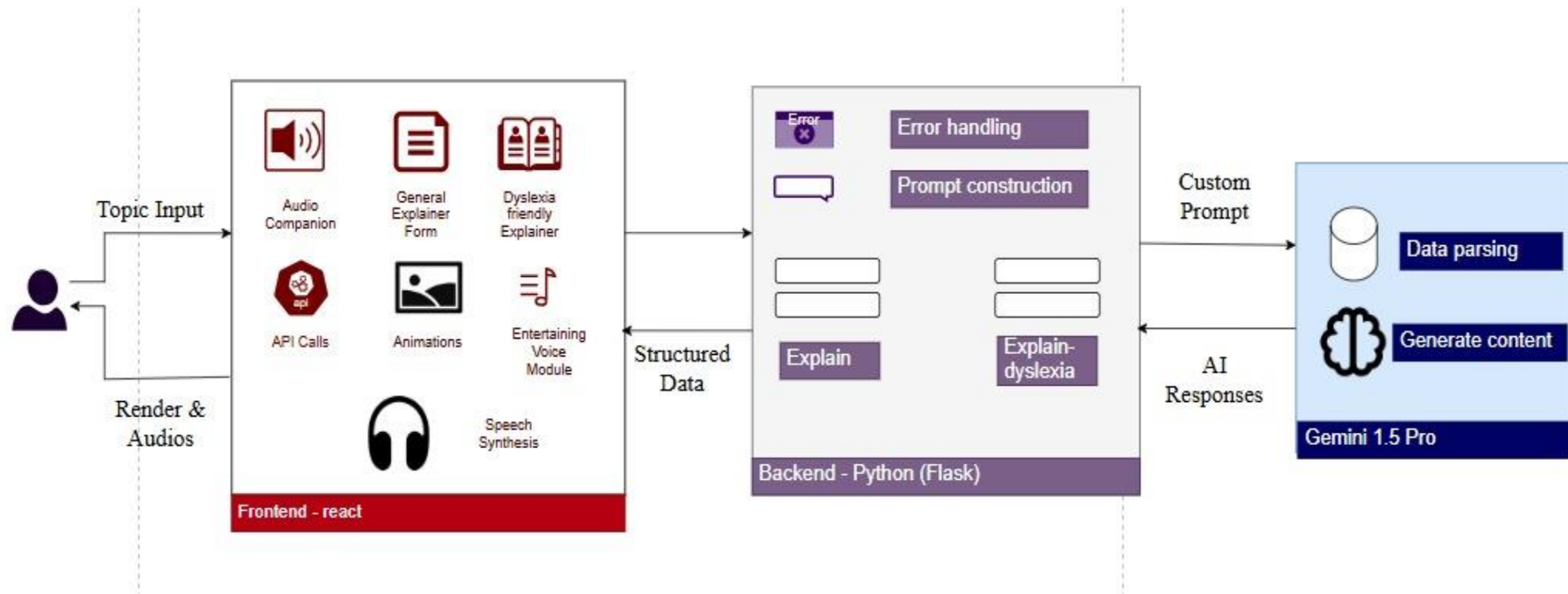
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Problem Statement

- Traditional educational methods offer uniform explanations for all learners.
- This one-size-fits-all approach leads to limited engagement and poor understanding.
- Learners of different age groups have varying comprehension needs.
- Most current tools lack adaptability to cater to these diverse needs.
- There is a need for an AI-powered solution that generates multi-level explanations based on cognitive ability.
- Such a tool can improve understanding, engagement, and retention across age groups.

Architecture Diagram



Solution Overview

- **User Input Flow:** Users enter a topic in the main input field.
- **Prompt Construction:** The backend dynamically builds a structured prompt using level-specific tags.
- **AI Communication:** Gemini 1.5 Pro processes the prompt and returns age-specific explanations.
- **Display Logic:** The frontend parses and animates these responses using React components and Framer Motion.
- **Error Handling:** Alert-based fallbacks for failed API calls.
- **Audio Support:** Users can optionally listen to explanations via **AudioCompanion** (age-specific tone/speed) or in a fun way using **EntertainingVoices** (e.g., wizard, comedian, RJ modes).
- **Accessibility:** A **DyslexiaPage** provides alternate explanation formats such as Simple, Steps, Q&A, and Story, enhancing inclusivity.

Tech Stack

Layer	Technology Used	Description
Frontend	React.js, Framer Motion	UI, animations, topic input form, response cards, audio support
Backend	Flask (Python)	Handles API requests, constructs prompts, parses AI output
AI Model	Gemini 1.5 Pro (Google)	Generates context-sensitive, multi-level educational content
Data Flow	REST API, JSON	Topic → Prompt → AI → Explanation → UI
Audio Layer	Text-to-Speech API (Browser or Plugin)	AudioCompanion with age-tone/speed + EntertainingVoices (wizard, comedian, RJ)
Accessibility	Conditional Rendering + Format Switches	DyslexiaPage with selectable formats: Simple, Steps, Q&A, Story
Security	HTTPS, CORS, Input Validation	Protects data exchange and ensures integrity

Implementation Details

Frontend (React)

- Users input a topic via text field.
- On clicking “Explain,” a POST request is triggered to the Flask backend.
- Responses are displayed in animated cards based on age group.
- If the API call fails, a popup `alert()` is triggered.
- **Audio Features:** AudioCompanion: Text-to-speech with tone/speed based on selected age group.
- EntertainingVoices: Adds fun character voices like wizard, comedian, and RJ for an engaging experience.
- **DyslexiaPage:** A dedicated view where users can select among four alternate reading formats for accessibility:
-  Simple,  Steps,  Q&A,  Story.

Backend (Flask) Receives topic and constructs structured prompt with custom tags ([KIDS], [TEENS], [EXPERTS]).

- Sends the prompt to Gemini 1.5 Pro via API and receives the multi-sectioned response.
- Uses regex to extract responses for each category.
- Returns a structured JSON containing content for:
- Explanation Cards (Kids, Teens, Experts)
- AudioCompanion playback
- DyslexiaPage formatted variations

Gemini AI Layer

- Gemini 1.5 Pro processes the prompt and provides contextualized, natural-language explanations for each target age group.
- The model's output is designed to maintain clarity, age-appropriateness, and tone consistency across all layers, including:
-  Audio narration (AudioCompanion, EntertainingVoices)
-  Alternate explanation structures (DyslexiaPage)

Demo Link

https://drive.google.com/file/d/1FPV-FA_LuTELV7lfULUyXxqFyGLQ87VF/view?usp=sharing

Thank you